LIBSORT – SYSTEM DEVELOPMENT REQUIREMENTS

Project Topic

Development of a Library Management System for Book Sorting

System Name

LibSort

Project Overview

LibSort is a library system designed to make it easier for students to find and borrow books and for library staff to know exactly where returned books should be placed.

The developer does not have direct access to the school's library or its existing database, so the system should initially be developed using the library structure and sample book data provided below.

The system should be designed so that the library information can easily be changed or expanded later.

Main Objectives

1. To develop a digital shelf map linking shelf codes to physical location.
2. To implement a scan-and-verify module that detects and alerts on misplaced books.
3. To evaluate the system through functional, usability, and performance testing.

1. Student Book Search

Students should be able to search for a book using:

- Book title
- Author
- ISBN/book ID
- Category/subject

The system should show:

- Book title
- Availability
- Shelf code
- Floor
- Section
- Exact location

Example:

Database System Concepts
Status: Available
Shelf Code: FF-CS-03
Location: First Floor → Art Section → Shelf 03

This allows students to easily find the book instead of searching through the library manually.

2. Returning and Sorting Books

When students return books, the books may be placed on a return table.

Library staff should be able to **search for or scan each returned book** and immediately see its registered location.

Example:

Book: Data Structures
Return Location: Top Floor → Science & Technology → Shelf 04

This helps staff return books to the correct shelves without having to remember or manually look for where each book belongs.

3. Scan-and-Verify

Each book can have a barcode or QR code.

When staff scan a book, LibSort should retrieve its assigned shelf and location.

The staff can then verify the shelf where the book is being returned.

If the selected/scanned shelf does not match the book's assigned shelf, the system should display an alert such as:

MISPLACED BOOK
This book belongs on Shelf TF-ST-04.

The system does **not** require electronic sensors or smart shelves. The verification is done through the scanning process.

4. Overdue Books

The system should record borrowed books and their due dates.

When a due date passes, the book should automatically be marked **Overdue**.

Staff should have an overdue section where they can see the affected books and borrowers.

There should also be an option for staff to send a reminder/notification to the student to return the overdue book, depending on the notification method the developer can practically implement.

5. Digital Shelf Map

The library currently has **three floors** with a total of **36 shelves**:

Floor	Section	Number of Shelves
Ground Floor	Social Sciences	9
First Floor	Art	10
Top Floor	Science & Technology	17
Total		**36**

The shelf codes can be structured clearly according to the floor and section.

For example:

- Ground Floor / Social Sciences: `GF-SS-01` to `GF-SS-09`
- First Floor / Art: `FF-ART-01` to `FF-ART-10`
- Top Floor / Science & Technology: `TF-ST-01` to `TF-ST-17`

These codes are examples of the naming structure and can be adjusted if necessary.

IMPORTANT: Locations Must Be Data

Shelf locations must be stored as database data, not permanently hard-coded into the application.

For example, `GF-SS-01` should be a database record containing information such as:

Floor: Ground Floor
Section: Social Sciences
Shelf: 01

The administrator should be able to **add, edit, remove, or change floors, sections, shelf codes and locations** without needing to rebuild the entire system.

This is important because the library's physical arrangement may change in the future.

It also means that the same LibSort application can be adapted for another library simply by changing the database/location records.

6. Initial Database

The developer should create a sample database containing enough records to demonstrate the system.

It should include:

- Books
- Authors
- Book IDs/ISBNs
- Floors
- Sections
- Shelf codes
- Book locations
- Available books
- Borrowed books
- Returned books
- Overdue books
- Sample students

The physical library structure should initially use the information currently available:

3 floors → 36 shelves → 3 sections.

The actual book records can be populated with sample/publicly available book data for development and testing.

Basic System Flow

Student:
Search book → Check availability → View exact location → Find book → Borrow

Staff:
Returned book → Search/scan book → System shows exact location → Return book to correct shelf

Overdue:
Due date passes → System marks book overdue → Staff sees alert → Staff sends reminder

Verification:
Scan book → System retrieves assigned shelf → Staff verifies location → Correct / Misplaced alert

The main focus of LibSort should remain **book location, easy book access, correct book sorting/returning, scan-and-verify, and overdue-book management**.